

Cross product

The Cross Product of two Vectors term $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ is:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

example:

$$\vec{a} = \langle 2, 3, 1 \rangle \quad \text{and} \quad \vec{b} = \langle -1, 2, 4 \rangle$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

You can then write it like this:

$$i[\quad] - j[\quad] + k[\quad]$$

- the sign alternates so it goes +, -, + this pattern will always be like this
Meaning you will always begin with a positive i and a -j and a positive k
- you still need to fill in the table you do this by dropping the row and column corresponding to the letter

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

- We have currently dropped i by covering the row i is in and column so the numbers that are highlighted in gray is what you place in the box for i

$$i \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} - j \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix} + k \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

- We are now going to find what j is so we drop the row and column the row j is in is dropped and column is also dropped so you copy down the numbers that aren't highlighted in gray

$$i \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} - j \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix} + k \begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 1 \\ -1 & 2 & 4 \end{vmatrix}$$

- We do the same thing for k we drop row and column and (the gray highlight is dropping) and copy down the numbers that are not highlighted

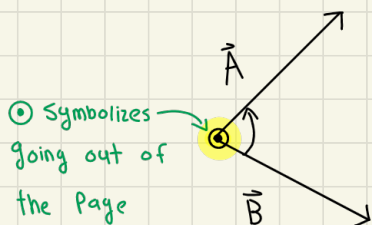
$$i \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} - j \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix} + k \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

- You then cross multiply and subtract

$$i[(3 \cdot 4) - (2 \cdot 1)] - j[(2 \cdot 4) - (-1 \cdot 1)] + k[(2 \cdot 2) - (-1 \cdot 3)]$$

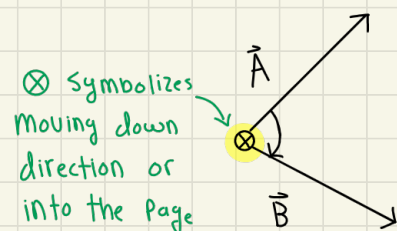
$$(10 - 2)i + (-9)j + 7k \Rightarrow 10i - 9j + 7k \text{ or } \langle 10, -9, 7 \rangle$$

★ answer will be a vector



$$\vec{A} \times \vec{B}$$

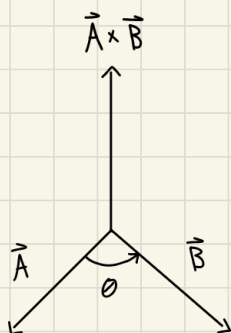
- You can use your right hand to find in the $\vec{A} \times \vec{B}$ is going inside or outside the page. We see the arrows tell us that it goes from $\vec{B}$ to $\vec{A}$. So you curl your right hand fingers in that direction. For this, $\vec{A} \times \vec{B}$ is going outside the page.



$$\vec{A} \times \vec{B}$$

- You can do the same thing with your right hand by curling your finger and seeing which your thumb goes. But this time you will curl your fingers from $\vec{A}$ to $\vec{B}$ and the direction $\vec{A} \times \vec{B}$ will be going down or into the page.

- This is what it looks like:



- another way to write the cross product is:

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

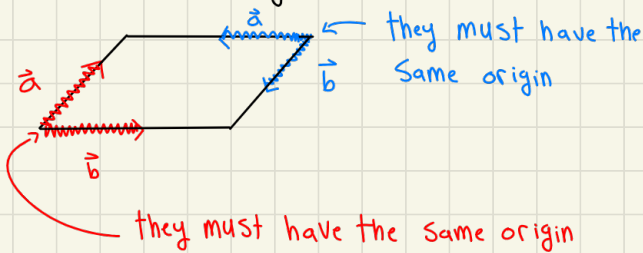
- two non-zero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if $\vec{a} \times \vec{b} = 0$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin(180)$$

$$|\vec{a} \times \vec{b}| = 0$$

- $\vec{a} \times \vec{b} = \text{Vector}$
- if $\vec{a} \times \vec{b} = 0 \Rightarrow \text{Parallel}$
- $|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$
- $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$

Applications: area of a Parallelogram



- you can use either one of these as long as they have the same origin

the area of the Parallelogram is $|\vec{A} \times \vec{B}|$ you take the cross product of $\vec{A}$ and $\vec{B}$ then take the magnitude

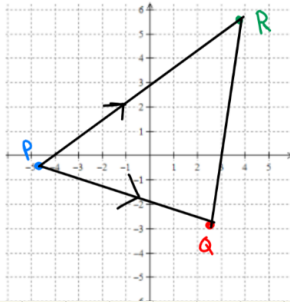
- triangle



the area of the triangle is $\frac{|\vec{a} \times \vec{b}|}{2}$
you take the cross product of $\vec{A}$ and $\vec{B}$ then take the magnitude and divide by 2

- they also need to have the same origin

5. Find a non-zero vector orthogonal to the plane through the points $P(-3,0)$, $Q(3,-2)$, $R(4,5)$ and find the area of the triangle.



$$\vec{PQ} = Q - P = \langle 6, -2, 0 \rangle$$

$$\vec{PR} = R - P = \langle 7, 5, 0 \rangle$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} i & j & k \\ 6 & -2 & 0 \\ 7 & 5 & 0 \end{vmatrix}$$

$$i \begin{vmatrix} -2 & 0 \\ 5 & 0 \end{vmatrix} - j \begin{vmatrix} 6 & 0 \\ 7 & 0 \end{vmatrix} + k \begin{vmatrix} 6 & -2 \\ 7 & 5 \end{vmatrix}$$

$$i(0-0) - j(0-0) + k(30 - (-14))$$

$$44k = (0, 0, 44) \Rightarrow \sqrt{0^2 + 0^2 + 44^2} = 44$$

• because you are looking for area of triangle

$$\frac{44}{2} = \boxed{22}$$

- Volume of Parallelepiped

$$|\vec{a} \cdot (\vec{b} \times \vec{c})|$$

after you have completed the cross product you are left with a dot product

Cross product do this first

6. Find the volume of the parallelopiped with adjacent edges PQ, PR and PS where $P(3, 0, 1), Q(-1, 2, 5), R(5, 1, -1), S(0, 4, 2)$

$$a = \vec{PQ} = Q - P = (-1, 2, 5) - (3, 0, 1) = \langle -4, 2, 4 \rangle$$

$$b = \vec{PR} = R - P = (5, 1, -1) - (3, 0, 1) = \langle 2, 1, -2 \rangle$$

$$c = \vec{PS} = S - P = (0, 4, 2) - (3, 0, 1) = \langle -3, 4, 1 \rangle$$

$$|\vec{a} \cdot (\vec{b} \times \vec{c})|$$

$$\vec{b} \times \vec{c} = \begin{vmatrix} i & j & k \\ 2 & 1 & -2 \\ -3 & 4 & 1 \end{vmatrix}$$

$$i \begin{vmatrix} 1 & -2 \\ 4 & 1 \end{vmatrix} - j \begin{vmatrix} 2 & -2 \\ -3 & 1 \end{vmatrix} + k \begin{vmatrix} 2 & 1 \\ -3 & 4 \end{vmatrix}$$

$$i[(1 \cdot 1) - (4 \cdot -2)] - j[(2 \cdot 1) - (-3 \cdot -2)] + k[(2 \cdot 4) - (-1 \cdot -3)]$$

• Setting to n
easier to plug into
equation

$$\vec{n} = 9i + 4j + 11k$$

$$a = \langle -4, 2, 4 \rangle \quad n = \langle 9, 4, 11 \rangle$$

take dot Product

$$a \cdot n = a_1 n_1 + a_2 n_2 + a_3 n_3$$

$$(-4)(9) + (2)(4) + (4)(11)$$

$$= \boxed{16}$$

- two unit vector that are orthogonal to both $\vec{a} = 3\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\vec{b} = 5\mathbf{i} + 9\mathbf{j} - 2\mathbf{k}$

 remember

$$\vec{A} \times \vec{B} = \langle -1, 0, 1 \rangle$$



$$\vec{A} \times \vec{B} = \langle 1, 0, 1 \rangle$$



$$\vec{a} \times \vec{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 3 \\ 5 & 9 & -2 \end{vmatrix}$$

$$\mathbf{i} \begin{vmatrix} -1 & 3 \\ 9 & -2 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 3 & 3 \\ 5 & -2 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 3 & -1 \\ 5 & 9 \end{vmatrix}$$

$$\mathbf{i} [(-2 \cdot -1) - (3 \cdot 9)] - \mathbf{j} [(3 \cdot -2) - (3 \cdot 5)] + \mathbf{k} [(3 \cdot 9) - (5 \cdot -1)]$$

$$-25\mathbf{i} + 21\mathbf{j} + 32\mathbf{k}$$

$$|\vec{a}| = \sqrt{(-25)^2 + (21)^2 + (32)^2} = 45.7$$

$$\textcircled{1} \quad \left\langle \frac{-25}{45.7}, \frac{21}{45.7}, \frac{32}{45.7} \right\rangle$$

$$\textcircled{2} \quad \left\langle \frac{25}{45.7}, -\frac{21}{45.7}, -\frac{32}{45.7} \right\rangle$$

Find the area of the parallelogram with vertices $A(1,0,2)$, $B(3,3,3)$, $C(7,5,8)$, $D(5,3,7)$

$$\vec{AB} = B - A = (3,3,3) - (1,0,2) = \langle 2, 3, 1 \rangle$$

$$\vec{AC} = C - A = (7,5,8) - (1,0,2) = \langle 6, 5, 6 \rangle$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 2 & 3 & 1 \\ 6 & 5 & 6 \end{vmatrix}$$

$$i \begin{vmatrix} 3 & 1 \\ 5 & 6 \end{vmatrix} - j \begin{vmatrix} 2 & 1 \\ 6 & 6 \end{vmatrix} + k \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix}$$

$$i[(6 \cdot 3) - (5 \cdot 1)] - j[(6 \cdot 2) - (6 \cdot 1)] + k[(6 \cdot 2) - (5 \cdot 3)]$$

$$13i - 6j - 3k \Rightarrow \sqrt{(13)^2 + (-6)^2 + (-3)^2}$$

$$= \sqrt{214}$$

• torque

- torque is a measurement of the rotational force

$$\vec{\tau} = \vec{r} \times \vec{F} \approx |\vec{r}| |\vec{F}| \sin \theta$$

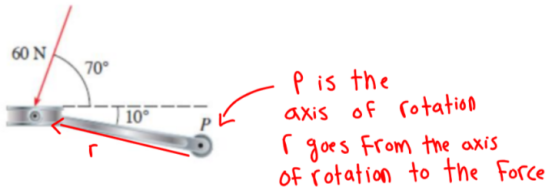
- this is a cross product

- $\vec{r}$ is the position vector (from the axis of rotation to the point where the force is applied)

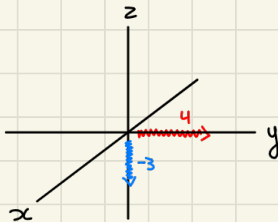
- $\vec{F}$ is the force vector

7. A 20 cm long wrench lies along the positive y axis and grips the bolt at the origin. A force is applied in the direction $(0, 4, -3)$ at the end of the wrench. Find the magnitude of force need to supply 100 N.m of torque to the bolt?

8. A bicycle pedal is pushed by a foot with a 60 - N forces as shown. The shaft of the pedal is 14 cm long. Find the magnitude of the torque about P. Enter the answer in N.m round your answer to one decimal place.



7) torque = $|\vec{F}| |\vec{r}| \sin \theta$

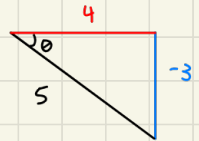


torque = 100 N.m

$r = 20 \text{ cm} \Rightarrow 0.2 \text{ m}$

$100 = |\vec{F}| (0.2) \left(-\frac{3}{5}\right)$

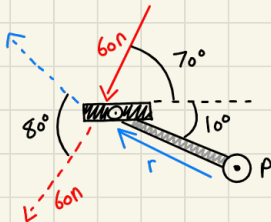
$|\vec{F}| = 833.3 \text{ N}$



$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{-3}{5}$

8)

You can move the $\vec{F}$ and $\vec{r}$ (Vectors can be moved around)



torque = $|\vec{F}| |\vec{r}| \sin \theta$

$(60) (0.14) \sin (80^\circ)$

$= 8.27 \text{ N.m}$